

ON THE HYPERBOLIC APPROXIMATION OF DIFFUSIVE EQUATIONS.

Firas Dhaouadi

Bordeaux INP, Institut de Mathématiques de Bordeaux, inria Bordeaux



July, 22nd 2026, Universidade de Santiago de Compostela

Outline

- 1 Introduction
- 2 Example 1: The Euler–Fourier system
- 3 Example 2: The Cahn–Hilliard equation
- 4 Conclusions

Acknowledgment to collaborators

This presentation is based on joint works with

- Michael Dumbser (University of Trento)
- Sergey Gavrilyuk (Aix-Marseille Université)

Ongoing work and extensions with

- Christian Rohde & Rahul Barthwal (University of Stuttgart)
- Mario Ricchiuto & Laura del Río Martín (Inria Bordeaux)
- Pierre Henri-Maire (CEA Cesta, Bordeaux)
- Alessia Lucca (University of Trento)
- Pep Mulet & Andreu Martorell (University of Valencia)

Diffusion equations in physics and mechanics

A large part of continuum physics is written in terms of parabolic operators:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x} \left(K(x, \theta) \frac{\partial \theta}{\partial x} \right) \quad (\text{Fourier's law of heat conduction})$$

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D(x, c) \frac{\partial c}{\partial x} \right) \quad (\text{Fick's law of mass diffusion})$$

$$\frac{\partial c}{\partial t} = \frac{\partial^2}{\partial x^2} \left(\Pi(c) - \gamma \frac{\partial^2 c}{\partial x^2} \right) \quad (\text{The Cahn–Hilliard equation})$$

$$\begin{cases} \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0 \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = \operatorname{div}(\mathbb{S}(\mathbf{u}, \nabla \mathbf{u})) \end{cases} \quad (\text{Navier–Stokes' system})$$

Diffusion equations in physics and mechanics

A large part of continuum physics is written in terms of parabolic operators:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x} \left(K(x, \theta) \frac{\partial \theta}{\partial x} \right) \quad (\text{Fourier's law of heat conduction})$$

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D(x, c) \frac{\partial c}{\partial x} \right) \quad (\text{Fick's law of mass diffusion})$$

$$\frac{\partial c}{\partial t} = \frac{\partial^2}{\partial x^2} \left(\Pi(c) - \gamma \frac{\partial^2 c}{\partial x^2} \right) \quad (\text{The Cahn–Hilliard equation})$$

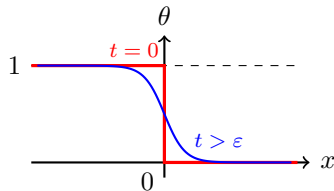
$$\begin{cases} \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0 \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = \operatorname{div}(\mathbb{S}(\mathbf{u}, \nabla \mathbf{u})) \end{cases} \quad (\text{Navier–Stokes' system})$$

- Require **postulating a constitutive law**: $\mathbf{q} = -K \nabla \theta$, $\mathbf{j} = -D \nabla c$, etc...
- The flux responds **instantaneously** to the gradient.

Instantaneous propagation – Paradox of the heat equation

$$\frac{\partial \theta}{\partial t} = K \Delta \theta, \quad \theta(x, 0) = \begin{cases} 1 & \text{if } x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\Rightarrow \theta(x, t) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{Kt}}\right).$$

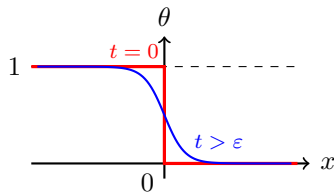


At every $t > 0$ the tails reach $x = \pm\infty$.

Instantaneous propagation – Paradox of the heat equation

$$\frac{\partial \theta}{\partial t} = K \Delta \theta, \quad \theta(x, 0) = \begin{cases} 1 & \text{if } x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\implies \theta(x, t) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{Kt}}\right).$$



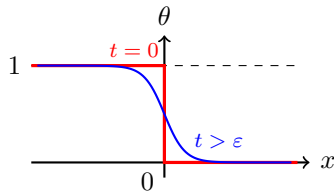
At every $t > 0$ the tails reach $x = \pm\infty$.

- For any $t > 0$, θ has **already** changed at **every** $x \Rightarrow$ **causality is violated**.

Instantaneous propagation – Paradox of the heat equation

$$\frac{\partial \theta}{\partial t} = K \Delta \theta, \quad \theta(x, 0) = \begin{cases} 1 & \text{if } x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\implies \theta(x, t) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{Kt}}\right).$$



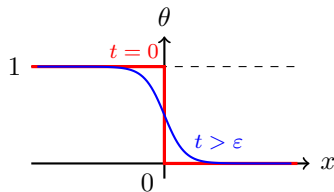
At every $t > 0$ the tails reach $x = \pm\infty$.

- For any $t > 0$, θ has **already** changed at **every** $x \Rightarrow$ **causality is violated**.
- **Incompatibility** with relativity (speed of light is the limit)
- Wrong physics at very short times / extreme conditions.

Instantaneous propagation – Paradox of the heat equation

$$\frac{\partial \theta}{\partial t} = K \Delta \theta, \quad \theta(x, 0) = \begin{cases} 1 & \text{if } x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\implies \theta(x, t) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{Kt}}\right).$$



At every $t > 0$ the tails reach $x = \pm\infty$.

- For any $t > 0$, θ has **already** changed at **every** $x \Rightarrow$ **causality is violated**.
- **Incompatibility** with relativity (speed of light is the limit)
- Wrong physics at very short times / extreme conditions.

[1] L. Tisza, "Transport phenomena in Helium II," *Nature*, vol. 141, p. 913, 1938.

[2] L. Landau, "On the Theory of Superfluidity of Helium II," *Journal of Physics USSR*, vol. 11, no. 1, pp. 91–92, 1947.

[3] Peshkov, V. P. "Second sound in helium II." *Sov. Phys. JETP* 11.3 (1960): 580-584.

The Cattaneo model of relativistic heat conduction

1) Take the classic linear heat equation

$$\frac{\partial \theta}{\partial t} - K \Delta \theta = 0 \quad \Longleftrightarrow \quad \begin{cases} \frac{\partial \theta}{\partial t} + \nabla \cdot \mathbf{q} = 0, \\ \mathbf{q} = -K \nabla \theta. \end{cases}$$

The Cattaneo model of relativistic heat conduction

1) Take the classic linear heat equation

$$\frac{\partial \theta}{\partial t} - K \Delta \theta = 0 \quad \Longleftrightarrow \quad \begin{cases} \frac{\partial \theta}{\partial t} + \nabla \cdot \mathbf{q} = 0, \\ \mathbf{q} = -K \nabla \theta. \end{cases}$$

2) Replace the **instantaneous** constitutive law by a *relaxed* version

$$\mathbf{q} = -K \nabla \theta \quad \Longrightarrow \quad \tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -K \nabla \theta, \quad 0 < \tau \ll 1.$$

The Cattaneo model of relativistic heat conduction

1) Take the classic linear heat equation

$$\frac{\partial \theta}{\partial t} - K \Delta \theta = 0 \quad \Longleftrightarrow \quad \begin{cases} \frac{\partial \theta}{\partial t} + \nabla \cdot \mathbf{q} = 0, \\ \mathbf{q} = -K \nabla \theta. \end{cases}$$

2) Replace the **instantaneous** constitutive law by a *relaxed* version

$$\mathbf{q} = -K \nabla \theta \quad \Longrightarrow \quad \tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -K \nabla \theta, \quad 0 < \tau \ll 1.$$

We obtain the system

$$\begin{aligned} \frac{\partial \theta}{\partial t} + \nabla \cdot \mathbf{q} &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \frac{K}{\tau} \nabla \theta &= -\frac{1}{\tau} \mathbf{q}. \end{aligned}$$

Old history - Copyright

The Cattaneo model is often referred to as the Maxwell-Cattaneo-Vernotte model.

- ① James Clerk Maxwell was the first to introduce such an approach for gases (1867):
- ② Cattaneo presented his model in 1948.
- ③ Paul Vernotte did it independently 10 years later.

PHYSIQUE MATHÉMATIQUE. — *Sur une forme de l'équation de la chaleur éliminant le paradoxe d'une propagation instantanée.* Note de M. **CARLO CATTANEO**, transmise par M. Joseph Kampé de Fériet.

Dans une Note récente ⁽¹⁾ M. Pierre Vernotte a proposé une correction de l'équation classique de la chaleur, qui, en changeant son caractère parabolique en hyperbolique, élimine le paradoxe d'une propagation instantanée. Or, dans un travail ⁽²⁾ publié en 1948, j'étais parvenu exactement aux mêmes corrections, en les justifiant par des considérations de théorie cinétique; voici les grandes lignes de ma démonstration.

La théorie (unidimensionnelle) de Fourier est fondée sur deux équations indéfinies

Properties of Cattaneo's model

- 1 The model is **strongly hyperbolic**. (with eigenvalues $\lambda_{1-4} = \pm\sqrt{K/\tau}, 0, 0$)

Properties of Cattaneo's model

- ① The model is **strongly hyperbolic**. (with eigenvalues $\lambda_{1-4} = \pm\sqrt{K/\tau}, 0, 0$)

Reminder: Hyperbolic PDE (1d case for simplicity)

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \cdot \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad \mathbf{Q} \in \mathbb{R}^n, \mathbf{A} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$$

is hyperbolic if $\mathbf{A}(\mathbf{Q})$ is diagonalizable in $\mathbb{R}$, for any admissible $\mathbf{Q} \in \mathbb{R}^n$.

Properties of Cattaneo's model

- ① The model is **strongly hyperbolic**. (with eigenvalues $\lambda_{1-4} = \pm\sqrt{K/\tau}, 0, 0$)

Reminder: Hyperbolic PDE (1d case for simplicity)

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \cdot \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad \mathbf{Q} \in \mathbb{R}^n, \mathbf{A} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$$

is hyperbolic if $\mathbf{A}(\mathbf{Q})$ is diagonalizable in $\mathbb{R}$, for any admissible $\mathbf{Q} \in \mathbb{R}^n$.

- ② The model dissipates a total energy

$$\frac{\partial}{\partial t} \int_{\Omega} \left(\frac{\theta^2}{2} + \frac{\tau}{2} \|\mathbf{q}\|^2 \right) d\Omega = - \int_{\Omega} \|\mathbf{q}\|^2 d\Omega \leq 0$$

Properties of Cattaneo's model

- ① The model is **strongly hyperbolic**. (with eigenvalues $\lambda_{1-4} = \pm\sqrt{K/\tau}, 0, 0$)

Reminder: Hyperbolic PDE (1d case for simplicity)

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \cdot \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad \mathbf{Q} \in \mathbb{R}^n, \mathbf{A} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$$

is hyperbolic if $\mathbf{A}(\mathbf{Q})$ is diagonalizable in $\mathbb{R}$, for any admissible $\mathbf{Q} \in \mathbb{R}^n$.

- ② The model dissipates a total energy

$$\frac{\partial}{\partial t} \int_{\Omega} \left(\frac{\theta^2}{2} + \frac{\tau}{2} \|\mathbf{q}\|^2 \right) d\Omega = - \int_{\Omega} \|\mathbf{q}\|^2 d\Omega \leq 0$$

- ③ smooth solutions are expected to converge to those of the heat equation in the limit $\tau \rightarrow 0$.

The notion of Hyperbolization

Given a general PDE

$$\frac{\partial \mathbf{Q}}{\partial t} = \mathcal{P}(\mathbf{Q}, \nabla \mathbf{Q}, \nabla^2 \mathbf{Q}, \dots)$$

find a family of **first-order** systems, parametrized by small ε ,

$$\frac{\partial \mathbf{Q}^\varepsilon}{\partial t} + \operatorname{div}(\mathbf{F}(\mathbf{Q}^\varepsilon)) = \frac{1}{\varepsilon} \mathbf{S}(\mathbf{Q}^\varepsilon),$$

such that

- The system is hyperbolic for all $\varepsilon > 0$,
- $\mathbf{Q}^\varepsilon \rightarrow \mathbf{Q}$ as $\varepsilon \rightarrow 0$ (convergence in the stiff relaxation limit);

The notion of Hyperbolization

Given a general PDE

$$\frac{\partial \mathbf{Q}}{\partial t} = \mathcal{P}(\mathbf{Q}, \nabla \mathbf{Q}, \nabla^2 \mathbf{Q}, \dots)$$

find a family of **first-order** systems, parametrized by small ε ,

$$\frac{\partial \mathbf{Q}^\varepsilon}{\partial t} + \operatorname{div}(\mathbf{F}(\mathbf{Q}^\varepsilon)) = \frac{1}{\varepsilon} \mathbf{S}(\mathbf{Q}^\varepsilon),$$

such that

- The system is hyperbolic for all $\varepsilon > 0$,
- $\mathbf{Q}^\varepsilon \rightarrow \mathbf{Q}$ as $\varepsilon \rightarrow 0$ (convergence in the stiff relaxation limit);
- **Consistency requirements:** Energy dissipation, Entropy production, conservation principles, variational structure, whenever applicable.

Two examples for the rest of this talk

1) Euler-Fourier system

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = \mathbf{0}$$

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + p \mathbf{u} - K \nabla \theta) = 0$$

- Compressible heat-conducting flows.

2) The Cahn–Hilliard equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c)$$

- Phase separation processes.
- 2nd and 4th order Diffusion.
- Non-convex energy.

Two examples for the rest of this talk

1) Euler-Fourier system

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = \mathbf{0}$$

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + p \mathbf{u} - K \nabla \theta) = 0$$

- Compressible heat-conducting flows.

2) The Cahn–Hilliard equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c)$$

- Phase separation processes.
- 2nd and 4th order Diffusion.
- Non-convex energy.

Two completely different strategies will be used.

Outline

- 1 Introduction
- 2 Example 1: The Euler–Fourier system
- 3 Example 2: The Cahn–Hilliard equation
- 4 Conclusions

Heat conduction in an inviscid compressible flow

Generally described by Euler equation + Fourier's law of heat conduction

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p(\rho, \eta) \mathbf{I}) = \mathbf{0},$$

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + p(\rho, \eta) \mathbf{u} - K \nabla \theta(\rho, \eta)) = 0.$$

Smooth solutions satisfy the Clausius-Duhem inequality

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div} \left(\rho \eta \mathbf{u} - \frac{K}{\theta(\rho, \eta)} \nabla \theta(\rho, \eta) \right) = \frac{K}{\theta(\rho, \eta)^2} \|\nabla \theta(\rho, \eta)\|^2 \geq 0.$$

Heat conduction in an inviscid compressible flow

Generally described by Euler equation + Fourier's law of heat conduction

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p(\rho, \eta) \mathbf{I}) = \mathbf{0},$$

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + p(\rho, \eta) \mathbf{u} - K \nabla \theta(\rho, \eta)) = 0.$$

Smooth solutions satisfy the Clausius-Duhem inequality

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div} \left(\rho \eta \mathbf{u} - \frac{K}{\theta(\rho, \eta)} \nabla \theta(\rho, \eta) \right) = \frac{K}{\theta(\rho, \eta)^2} \|\nabla \theta(\rho, \eta)\|^2 \geq 0.$$

$\implies$ *Let us make it hyperbolic!*

How about applying Cattaneo ?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbb{I}) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \mathbf{u} + p \mathbf{u} + \mathbf{q}) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{K}{\tau} \nabla \theta = -\frac{1}{\tau} \mathbf{q}$$

How about applying Cattaneo ?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbb{I}) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \mathbf{u} + p \mathbf{u} + \mathbf{q}) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{K}{\tau} \nabla \theta = -\frac{1}{\tau} \mathbf{q}$$

No Galilean invariance

How about applying Cattaneo ?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbb{I}) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \mathbf{u} + p \mathbf{u} + \mathbf{q}) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{q}}{\partial \mathbf{x}} \mathbf{u} + \frac{K}{\tau} \nabla \theta = -\frac{1}{\tau} \mathbf{q}$$

No Galilean invariance (fixed by C.I. Christov, 2009).

How about applying Cattaneo ?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbb{I}) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \mathbf{u} + p \mathbf{u} + \mathbf{q}) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{q}}{\partial \mathbf{x}} \mathbf{u} + \frac{K}{\tau} \nabla \theta = -\frac{1}{\tau} \mathbf{q}$$

No Galilean invariance (fixed by C.I. Christov, 2009).

No Entropy inequality: The Gibbs identity implies that

$$\frac{\partial \rho \eta}{\partial t} + \nabla \cdot \left(\rho \eta \mathbf{u} + \frac{\mathbf{q}}{\theta} \right) = -\frac{\mathbf{q} \cdot \nabla \theta}{\theta^2} \stackrel{?}{>} 0$$

How about applying Cattaneo ?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbb{I}) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \mathbf{u} + p \mathbf{u} + \mathbf{q}) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{q}}{\partial \mathbf{x}} \mathbf{u} + \frac{K}{\tau} \nabla \theta = -\frac{1}{\tau} \mathbf{q}$$

No Galilean invariance (fixed by C.I. Christov, 2009).

No Entropy inequality: The Gibbs identity implies that

$$\frac{\partial \rho \eta}{\partial t} + \nabla \cdot \left(\rho \eta \mathbf{u} + \frac{\mathbf{q}}{\theta} \right) = -\frac{\mathbf{q} \cdot \nabla \theta}{\theta^2} \stackrel{?}{>} 0$$

not necessarily satisfied because $\mathbf{q} \neq -K \nabla \theta \quad \forall \tau > 0$.

How about applying Cattaneo ?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbb{I}) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \mathbf{u} + p \mathbf{u} + \mathbf{q}) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \frac{\partial \mathbf{q}}{\partial \mathbf{x}} \mathbf{u} + \frac{K}{\tau} \nabla \theta = -\frac{1}{\tau} \mathbf{q}$$

No Galilean invariance (fixed by C.I. Christov, 2009).

No Entropy inequality: The Gibbs identity implies that

$$\frac{\partial \rho \eta}{\partial t} + \nabla \cdot \left(\rho \eta \mathbf{u} + \frac{\mathbf{q}}{\theta} \right) = -\frac{\mathbf{q} \cdot \nabla \theta}{\theta^2} \stackrel{?}{>} 0$$

not necessarily satisfied because $\mathbf{q} \neq -K \nabla \theta \quad \forall \tau > 0$.

Look deeper into the literature

1. Cimmelli VA, Kosinski W, Saxton K. 1991 A new approach to the theory of heat conduction with finite wave speeds. *Le Matematiche* 46, 95–105.
2. Rubin M. 1992 Hyperbolic heat conduction and the second law. *Int. J. Eng. Sci.* 30, 1665–1676.
3. Ganghoffer, J. F., Do, X. N., & Fumeron, S. (2025). Emergence of anomalous hyperbolic heat conduction in homogenized media. *Composite Structures*, 119492.
4. Angeles F. 2022 **Non-hyperbolicity** of the inviscid Cattaneo–Christov system for compressible fluid flow in several space dimensions. *Quart. J. Mech. Appl. Math.* 75, 147–170.
5. **Other models in the literature:** Guyer-Krumhansl model, Kinetic theory, works of Ruggeri-Müller, Godunov-Peshkov-Romenski line,... etc .

Consistency requirements

- ① Asymptotically consistent with Fourier's law.
- ② First-order hyperbolic system with an explicit eigenstructure
- ③ Total energy is adequately dissipated.
- ④ Entropy production is positive (2nd thermodynamics principle)

Consistency requirements

- ① Asymptotically consistent with Fourier's law.
- ② First-order hyperbolic system with an explicit eigenstructure
- ③ Total energy is adequately dissipated.
- ④ Entropy production is positive (2nd thermodynamics principle)
- ⑤ Galilean invariant.

Consistency requirements

- ① Asymptotically consistent with Fourier's law.
- ② First-order hyperbolic system with an explicit eigenstructure
- ③ Total energy is adequately dissipated.
- ④ Entropy production is positive (2nd thermodynamics principle)
- ⑤ Galilean invariant.
- ⑥ **Conservative part can be derived from a variational principle**

Euler–Lagrange equations

Least action: nature picks the field making the action $a = \iint \mathcal{L} dx dt$ stationary ($\delta a = 0$). For a field $u(\mathbf{x}, t)$,

$$\mathcal{L}(u, u_t, \nabla u) \implies \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial u_t} \right) + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial \nabla u} \right) = \frac{\partial \mathcal{L}}{\partial u}.$$

Euler–Lagrange equations

Least action: nature picks the field making the action $a = \iint \mathcal{L} dx dt$ stationary ($\delta a = 0$). For a field $u(\mathbf{x}, t)$,

$$\mathcal{L}(u, u_t, \nabla u) \implies \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial u_t} \right) + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial \nabla u} \right) = \frac{\partial \mathcal{L}}{\partial u}.$$

Simplest example: a vibrating string

$$\mathcal{L} = \underbrace{\frac{1}{2} u_t^2}_{\text{kinetic}} - \underbrace{\frac{1}{2} \|\nabla u\|^2}_{\text{potential}} \implies \frac{\partial \mathcal{L}}{\partial u_t} = \frac{\partial u}{\partial t}, \quad \frac{\partial \mathcal{L}}{\partial u_x} = -\nabla u, \quad \frac{\partial \mathcal{L}}{\partial u} = 0.$$

Euler–Lagrange equations

Least action: nature picks the field making the action $a = \iint \mathcal{L} dx dt$ stationary ($\delta a = 0$). For a field $u(\mathbf{x}, t)$,

$$\mathcal{L}(u, u_t, \nabla u) \implies \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial u_t} \right) + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial \nabla u} \right) = \frac{\partial \mathcal{L}}{\partial u}.$$

Simplest example: a vibrating string

$$\mathcal{L} = \underbrace{\frac{1}{2} u_t^2}_{\text{kinetic}} - \underbrace{\frac{1}{2} \|\nabla u\|^2}_{\text{potential}} \implies \frac{\partial \mathcal{L}}{\partial u_t} = \frac{\partial u}{\partial t}, \quad \frac{\partial \mathcal{L}}{\partial u_x} = -\nabla u, \quad \frac{\partial \mathcal{L}}{\partial u} = 0.$$

We get:

$$u_{tt} - \Delta u = 0.$$

Euler–Lagrange equations: The case of continua

For a fluid, the Lagrangian is again kinetic minus internal energy,

$$\mathcal{L}(\rho, \mathbf{u}) = \int_{\Omega_t} \left(\underbrace{\frac{1}{2}\rho \|\mathbf{u}\|^2}_{\text{kinetic}} - \underbrace{\rho \varepsilon(\rho, \eta)}_{\text{internal energy}} \right) d\Omega.$$

Euler–Lagrange equations: The case of continua

For a fluid, the Lagrangian is again kinetic minus internal energy,

$$\mathcal{L}(\rho, \mathbf{u}) = \int_{\Omega_t} \left(\underbrace{\frac{1}{2}\rho \|\mathbf{u}\|^2}_{\text{kinetic}} - \underbrace{\rho \varepsilon(\rho, \eta)}_{\text{internal energy}} \right) d\Omega.$$

But we cannot vary ρ and $\mathbf{u}$ independently: they are linked by mass conservation. A displacement $\delta \mathbf{x}$ of the fluid particles constrains both fields,

$$\delta \rho = -\operatorname{div}(\rho \delta \mathbf{x}), \quad \delta \mathbf{u} = \frac{\partial \delta \mathbf{x}}{\partial t} + \frac{\partial \delta \mathbf{x}}{\partial \mathbf{x}} \mathbf{u} - \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \delta \mathbf{x}.$$

Euler–Lagrange equations: The case of continua

For a fluid, the Lagrangian is again kinetic minus internal energy,

$$\mathcal{L}(\rho, \mathbf{u}) = \int_{\Omega_t} \left(\underbrace{\frac{1}{2}\rho \|\mathbf{u}\|^2}_{\text{kinetic}} - \underbrace{\rho \varepsilon(\rho, \eta)}_{\text{internal energy}} \right) d\Omega.$$

But we cannot vary ρ and $\mathbf{u}$ independently: they are linked by mass conservation. A displacement $\delta \mathbf{x}$ of the fluid particles constrains both fields,

$$\delta \rho = -\operatorname{div}(\rho \delta \mathbf{x}), \quad \delta \mathbf{u} = \frac{\partial \delta \mathbf{x}}{\partial t} + \frac{\partial \delta \mathbf{x}}{\partial \mathbf{x}} \mathbf{u} - \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \delta \mathbf{x}.$$

Only $\delta \mathbf{x}$ is free. Injecting these *constrained* variations into $\delta a = 0$ and integrating by parts:

$$\implies \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} \left(\rho \mathbf{u} \otimes \mathbf{u} + \rho^2 \frac{\partial \varepsilon}{\partial \rho} \mathbf{I} \right) = 0$$

Thermal displacement (Green-Naghdi 1991)

In this paper:

[1] Green, A. E., & Naghdi, P. (1991). A re-examination of the basic postulates of thermomechanics. *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences*, 432(1885), 171-194.

The authors introduce an independent auxiliary potential $\phi(\mathbf{x}, t)$ as a thermal analogue of the kinematic variables such that

$$\dot{\phi}(\mathbf{x}, t) = -\theta(\mathbf{x}, t)$$

One can then write the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) \right) d\Omega,$$

where

$$\varepsilon(\rho, \eta) = \varepsilon^*(\rho, \dot{\phi}) - \eta \dot{\phi}, \quad \text{with} \quad \eta = \frac{\partial \varepsilon^*}{\partial \dot{\phi}}.$$

Entropy as an Euler-Lagrange equation

Given the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) \right) d\Omega, \quad \left(\dot{\phi} = \frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi \right)$$

Entropy as an Euler-Lagrange equation

Given the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) \right) d\Omega, \quad \left(\dot{\phi} = \frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi \right)$$

One obtains (Recall that $\eta = \frac{\partial \varepsilon^*}{\partial \dot{\phi}}$.)

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} \left(\rho \mathbf{u} \otimes \mathbf{u} + \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} \mathbf{I} \right) = 0, \quad (\text{Euler-Lagrange for } \delta \mathbf{x})$$

$$\frac{\partial}{\partial t} \left(\rho \frac{\partial \varepsilon^*}{\partial \dot{\phi}} \right) + \operatorname{div} \left(\rho \frac{\partial \varepsilon^*}{\partial \dot{\phi}} \mathbf{u} \right) = 0, \quad (\text{Euler-Lagrange for } \delta \phi)$$

Entropy as an Euler-Lagrange equation

Given the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) \right) d\Omega, \quad \left(\dot{\phi} = \frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi \right)$$

One obtains (Recall that $\eta = \frac{\partial \varepsilon^*}{\partial \dot{\phi}}$.)

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} \left(\rho \mathbf{u} \otimes \mathbf{u} + \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} \mathbf{I} \right) = 0, \quad (\text{Euler-Lagrange for } \delta \mathbf{x})$$

$$\frac{\partial}{\partial t} (\rho \eta) + \operatorname{div} (\rho \eta \mathbf{u}) = 0, \quad (\text{Euler-Lagrange for } \delta \phi)$$

$$\frac{\partial \rho}{\partial t} + \operatorname{div} (\rho \mathbf{u}) = 0 \quad (\text{Constraint})$$

- A similar idea was also used in Lagrangian coordinates in *Peshkov, Pavelka, Grmela and Romenski (2018)*.

Extension of Green-Naghdi's philosophy

Consider the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \nabla \phi, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^{\star}(\rho, \dot{\phi}) - \frac{1}{2} \alpha(\rho) \|\nabla \phi\|^2 \right) d\Omega,$$

where the function $\alpha(\rho)$ is an arbitrary positive function of density.

Extension of Green-Naghdi's philosophy

Consider the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \nabla \phi, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) - \frac{1}{2} \alpha(\rho) \|\nabla \phi\|^2 \right) d\Omega,$$

where the function $\alpha(\rho)$ is an arbitrary positive function of density.

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + P \mathbf{I} + \alpha(\rho) \nabla \phi \otimes \nabla \phi) = 0,$$

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \nabla \phi) = 0,$$

where $P = \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} + \frac{1}{2}(\rho \alpha'(\rho) - \alpha(\rho)) \|\nabla \phi\|^2$

Extension of Green-Naghdi's philosophy

Consider the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \nabla \phi, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) - \frac{1}{2} \alpha(\rho) \|\nabla \phi\|^2 \right) d\Omega,$$

where the function $\alpha(\rho)$ is an arbitrary positive function of density.

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + P \mathbf{I} + \alpha(\rho) \nabla \phi \otimes \nabla \phi) = 0,$$

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \nabla \phi) = 0,$$

where $P = \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} + \frac{1}{2}(\rho \alpha'(\rho) - \alpha(\rho)) \|\nabla \phi\|^2$

✗ The PDE is of second order and depends on $\nabla \phi$.

Solution: First-order reduction

Let us introduce a new variable $\mathbf{j}$ defined as

$$\mathbf{j} = \nabla \phi$$

We then take the material derivative of this equation

$$\mathcal{D}_t \mathbf{j} = \nabla \phi_t + \frac{\partial \nabla \phi}{\partial \mathbf{x}} \cdot \mathbf{u}$$

which yields

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla (\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} = 0$$

Dissipationless system of equations

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + \Pi) = 0, \quad \Pi = P(\rho, \eta, \mathbf{j}) \mathbf{I} + \alpha(\rho) \mathbf{j} \otimes \mathbf{j}$$

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla(\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} = 0,$$

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \mathbf{j}) = 0.$$

Dissipationless system of equations

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + \Pi) = 0, \quad \Pi = P(\rho, \eta, \mathbf{j}) \mathbf{I} + \alpha(\rho) \mathbf{j} \otimes \mathbf{j}$$

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla(\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} = 0,$$

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \mathbf{j}) = 0.$$

Total energy conservation is obtained as a consequence

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + \Pi \mathbf{u} + \mathbf{q}) = 0, \quad \mathbf{q} = \alpha(\rho) \theta(\rho, \eta) \mathbf{j}$$

Dissipationless system of equations

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + \Pi) = 0, \quad \Pi = P(\rho, \eta, \mathbf{j}) \mathbf{I} + \alpha(\rho) \mathbf{j} \otimes \mathbf{j}$$

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla(\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} = 0,$$

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \mathbf{j}) = 0.$$

Total energy conservation is obtained as a consequence

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + \Pi \mathbf{u} + \mathbf{q}) = 0, \quad \mathbf{q} = \alpha(\rho) \theta(\rho, \eta) \mathbf{j}$$

Additional term in the energy conservation *should* be the heat flux.

Rayleigh dissipation function

$$\begin{aligned}
 \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) &= 0, \\
 \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + P(\rho, \eta, \mathbf{j}) \mathbf{I} + \alpha(\rho) \mathbf{j} \otimes \mathbf{j}) &= 0, \\
 \frac{\partial \mathbf{j}}{\partial t} + \nabla(\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} &= -\frac{\partial \mathcal{R}}{\partial \mathbf{j}}, \\
 \frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \mathbf{j}) &= \frac{\alpha(\rho)}{\theta(\rho, \eta)} \frac{\partial \mathcal{R}}{\partial \mathbf{j}} \cdot \mathbf{j}.
 \end{aligned}$$

Here $\mathcal{R}$ is the *Rayleigh dissipation* function and which we take in the simplest form as

$$\mathcal{R} = \frac{1}{2\tau} \|\mathbf{j}\|^2, \quad \frac{\partial \mathcal{R}}{\partial \mathbf{j}} = \frac{1}{\tau} \mathbf{j}$$

Asymptotic analysis: compatibility with Fourier's law

Expand every variable in powers of the relaxation time τ ,

$$\rho = \rho_0 + O(\tau), \quad \mathbf{u} = \mathbf{u}_0 + O(\tau), \quad \eta = \eta_0 + O(\tau), \quad \mathbf{j} = \mathbf{j}_0 + \tau \mathbf{j}_1 + o(\tau),$$

and substitute into the $\mathbf{j}$ equation. Matching orders in τ gives

$$\mathbf{j}_0 = 0, \quad \mathbf{j}_1 = -\nabla \theta(\rho_0, \eta_0) \quad \Longrightarrow \quad \boxed{\mathbf{j} = -\tau \nabla \theta(\rho_0, \eta_0) + o(\tau).}$$

Asymptotic analysis: compatibility with Fourier's law

Expand every variable in powers of the relaxation time τ ,

$$\rho = \rho_0 + O(\tau), \quad \mathbf{u} = \mathbf{u}_0 + O(\tau), \quad \eta = \eta_0 + O(\tau), \quad \mathbf{j} = \mathbf{j}_0 + \tau \mathbf{j}_1 + o(\tau),$$

and substitute into the $\mathbf{j}$ equation. Matching orders in τ gives

$$\mathbf{j}_0 = 0, \quad \mathbf{j}_1 = -\nabla \theta(\rho_0, \eta_0) \quad \Longrightarrow \quad \boxed{\mathbf{j} = -\tau \nabla \theta(\rho_0, \eta_0) + o(\tau).}$$

The heat flux vector then reads

$$\mathbf{q} = -\tau \alpha(\rho_0) \theta(\rho_0, \eta_0) \nabla \theta(\rho_0, \eta_0),$$

Asymptotic analysis: compatibility with Fourier's law

Expand every variable in powers of the relaxation time τ ,

$$\rho = \rho_0 + O(\tau), \quad \mathbf{u} = \mathbf{u}_0 + O(\tau), \quad \eta = \eta_0 + O(\tau), \quad \mathbf{j} = \mathbf{j}_0 + \tau \mathbf{j}_1 + o(\tau),$$

and substitute into the $\mathbf{j}$ equation. Matching orders in τ gives

$$\mathbf{j}_0 = 0, \quad \mathbf{j}_1 = -\nabla \theta(\rho_0, \eta_0) \quad \Longrightarrow \quad \boxed{\mathbf{j} = -\tau \nabla \theta(\rho_0, \eta_0) + o(\tau).}$$

The heat flux vector then reads

$$\mathbf{q} = -\tau \alpha(\rho_0) \theta(\rho_0, \eta_0) \nabla \theta(\rho_0, \eta_0),$$

which is compatible with **Fourier's law** $\mathbf{q} = -K \nabla \theta$, provided

$$\boxed{\tau = \frac{K}{\alpha(\rho_0) \theta(\rho_0, \eta_0)}}.$$

Energy convexity

Total energy is given by

$$E(\rho, \mathbf{m}, s, \mathbf{j}) = \frac{1}{2\rho} \|\mathbf{m}\|^2 + \rho\varepsilon(\rho, s/\rho) + \frac{1}{2}\alpha(\rho) \|\mathbf{j}\|^2, \quad \mathbf{m} = \rho\mathbf{u}, s = \rho\eta$$

Sufficient criterion for energy convexity

$$\text{if } \frac{\partial^2}{\partial \rho^2} \left(\frac{1}{\alpha(\rho)} \right) \leq 0, \quad \text{for } \rho > 0.$$

then E is also a convex function of $\mathbf{Q}$.

We choose a simple function fitting this criterion

$$\alpha(\rho) = \frac{\varkappa^2}{\rho}, \quad \varkappa = \text{cst.}$$

Energy convexity

Total energy is given by

$$E(\rho, \mathbf{m}, s, \mathbf{j}) = \frac{1}{2\rho} \|\mathbf{m}\|^2 + \rho\varepsilon(\rho, s/\rho) + \frac{1}{2}\alpha(\rho) \|\mathbf{j}\|^2, \quad \mathbf{m} = \rho\mathbf{u}, s = \rho\eta$$

Sufficient criterion for energy convexity

$$\text{if } \frac{\partial^2}{\partial \rho^2} \left(\frac{1}{\alpha(\rho)} \right) \leq 0, \quad \text{for } \rho > 0.$$

then E is also a convex function of $\mathbf{Q}$.

We choose a simple function fitting this criterion

$$\alpha(\rho) = \frac{\varkappa^2}{\rho}, \quad \varkappa = cst.$$

(Another possibility is $\alpha(\rho) = cst$, taken in *Peshkov et.al. (2018)*)

Hyperbolicity

system can be cast into quasilinear form

$$\frac{\partial \mathbf{V}}{\partial t} + \mathbf{A}(\mathbf{V}) \frac{\partial \mathbf{V}}{\partial \mathbf{x}} = 0$$

where $\mathbf{A}$ admits 8 eigenvalues whose expressions are given by

$$\left\{ \begin{array}{l} \chi_1 = u_1 - \sqrt{Z_1 + Z_2}, \\ \chi_2 = u_1 - \sqrt{Z_1 - Z_2}, \\ \chi_{3-6} = u_1, \\ \chi_7 = u_1 + \sqrt{Z_1 - Z_2}, \\ \chi_8 = u_1 + \sqrt{Z_1 + Z_2} \end{array} \right. \quad \text{where} \quad \left\{ \begin{array}{l} Z_1 = \frac{1}{2} (a_p^2 + a_T^2 + a_j^2), \\ Z_2 = \sqrt{a_{pT}^4 + \frac{1}{4} (a_p^2 - a_T^2)^2}, \\ a_p^2 = \frac{\partial p}{\partial \rho}, \quad a_T^2 = \frac{\kappa^2}{\rho^2} \frac{\partial \theta}{\partial \eta}, \\ a_{pT}^4 = \frac{\kappa^2}{\rho^2} \frac{\partial p}{\partial \eta} \frac{\partial \theta}{\partial \rho}, \quad a_j^2 = \frac{2\kappa^2}{\rho^2} (j_2^2 + j_3^2). \end{array} \right.$$

Riemann problem: A summary of results

Studying the Rankine-Hugoniot conditions yields

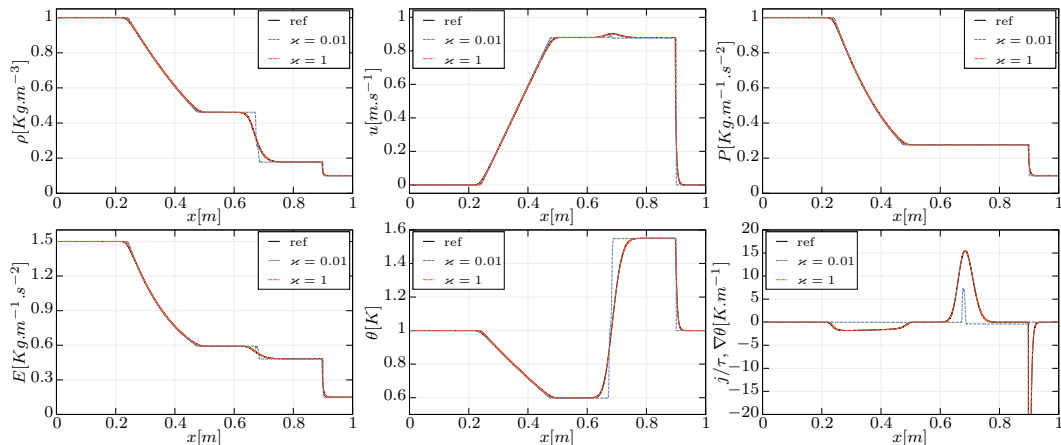
- **No contact discontinuities.** The Euler contact discontinuity is replaced by two new waves.
- **Nonclassical shocks:** The model admits admissible *expansion shocks*, *compression fans* and *compound shocks*.
- A Clausius–Duhem inequality

$$\left[\rho(u - D)\eta + \frac{\kappa^2}{\rho} j \right] \geq 0.$$

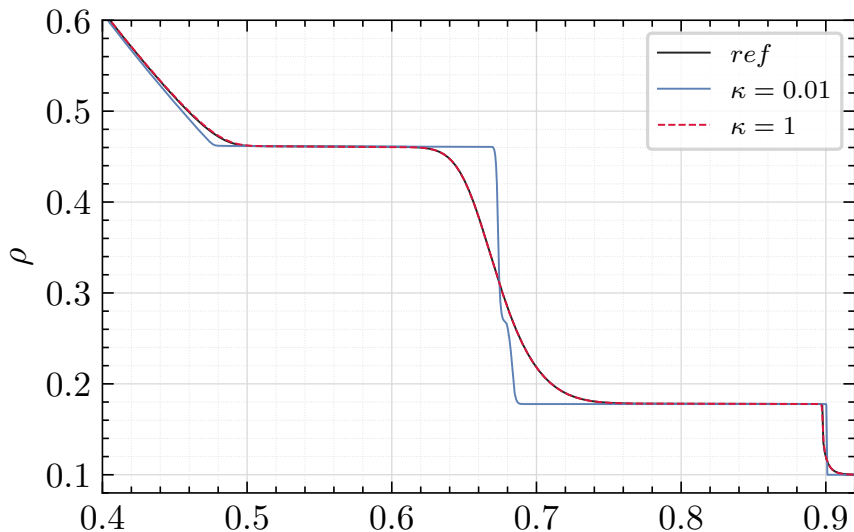
acts as the admissibility criterion.

(D is the front speed.)

Recovery of Fourier law: Shock tube problem (MUSCL-Hancock)



Recovery of Fourier law: Shock tube problem - Zoom



Expansion shock solution (No source term)

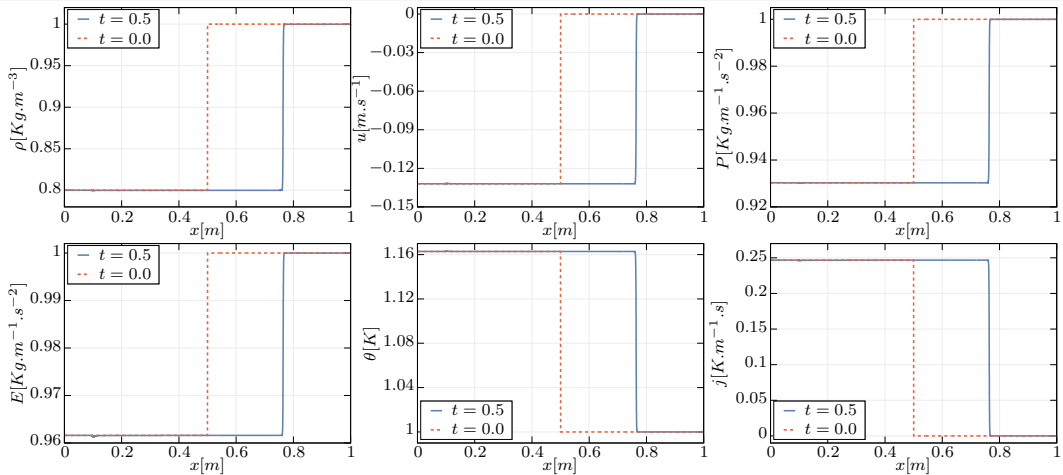


Figure 2: Numerical result for an expansion shock solution on the computational domain $[0, 1]$, discretized over $N = 10000$ cells displayed at final time $t = 0.5$. Parameters: $CFL = 0.9$,

Multi-D result (By W.Boscheri [JSC, 2026])

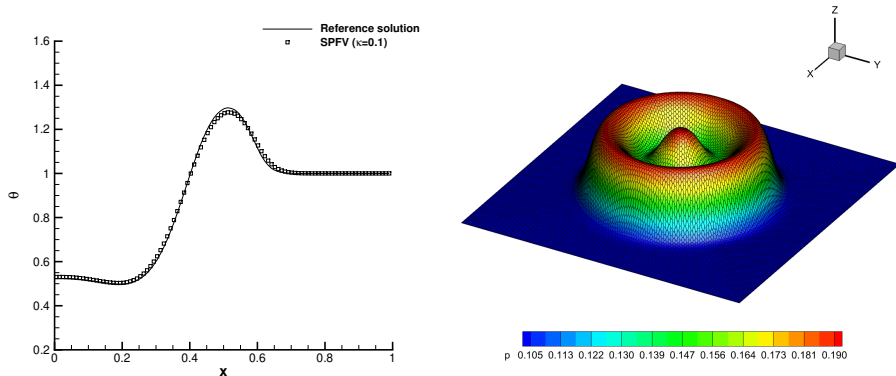


Figure 3: Explosion problem at time $t = 0.2$ with 200 equidistant points. Comparison between the Euler–Fourier solution and the hyperbolic model.

Multi-D result (By W.Boscheri [JSC, 2026])

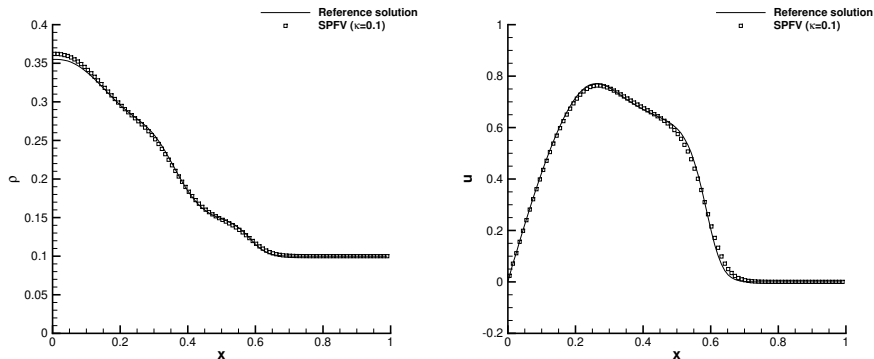


Figure 4: Explosion problem at time $t = 0.2$ with 200 equidistant points is used for the SPFV solution. 3D view, Voronoi mesh and color map of the pressure distribution are shown in the bottom right panel.

Outline

- 1 Introduction
- 2 Example 1: The Euler–Fourier system
- 3 Example 2: The Cahn–Hilliard equation
- 4 Conclusions

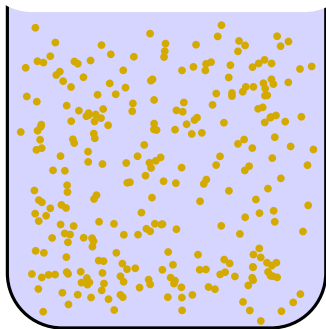
Cahn-Hilliard equations (1958)

The Cahn-Hilliard equation writes

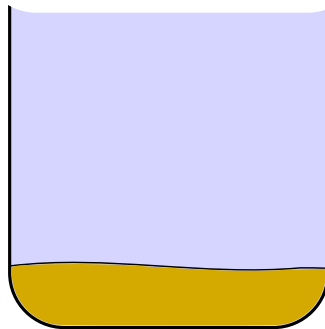
$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

- $c \in [-1, 1]$ is the order parameter indicating the phases.
- $\gamma \ll 1$ is such that $\sqrt{\gamma}$ is the diffuse interface characteristic length.
- describes phase separation in binary systems: spinodal decomposition, Ostwald Ripening phenomena, etc.

Phase separation: example of sedimentation



Dispersed phase



Separated phases

Conservative form and chemical potential

The Cahn-Hilliard equation can be cast into a conservation-law form which writes

$$\frac{\partial c}{\partial t} + \operatorname{div}(\mathbf{j}) = 0,$$

where the mass flux $\mathbf{j}$ is **assumed** to obey a generalized Fick's law such that

$$\mathbf{j} = -\nabla\mu,$$

and μ is the chemical potential of the system given by

$$\mu = \frac{\delta f}{\delta c} = \frac{\partial f}{\partial c} - \operatorname{div} \left(\frac{\partial f}{\partial \nabla c} \right) = c^3 - c - \gamma \Delta c,$$

where

$$f(c, \nabla c) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2,$$

Lyapunov functional

CH equation admits the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

Indeed, we have

$$\frac{\partial f}{\partial t} + \operatorname{div}(\mathcal{H}) = -\|\nabla \mu\|^2, \quad \mathcal{H} = -\mu \nabla \mu - \frac{\partial f}{\partial \nabla c} \Delta \mu,$$

which in integral form writes

$$\frac{\partial F}{\partial t} = - \int_{\mathcal{D}} \|\nabla \mu\|^2 \, d\Omega \leq 0.$$

Modified action functional

Based on the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

let us introduce the following action functional

$$a = \int_t \int_{\mathcal{D}} \mathcal{L} \, d\mathcal{D} dt$$

where

$$\mathcal{L} \left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t} \right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2.$$

- φ is a new variable substituting c as the order parameter.
- $\alpha \gg 1$ so that $(c - \varphi)$ vanishes in the limit $\alpha \rightarrow +\infty$.
- $\beta \ll 1$ is a small parameter.

On penalty methods

Idea: replace a hard constraint by a soft penalty that punishes its violation — the larger the penalty weight α , the closer the constraint is enforced.

$$\min_u E(u) \quad \text{s.t.} \quad u = g \quad \longrightarrow \quad \min_u E(u) + \frac{\alpha}{2} \|u - g\|^2.$$

On penalty methods

Idea: replace a hard constraint by a soft penalty that punishes its violation — the larger the penalty weight α , the closer the constraint is enforced.

$$\min_u E(u) \text{ s.t. } u = g \quad \longrightarrow \quad \min_u E(u) + \frac{\alpha}{2} \|u - g\|^2.$$

Example: image denoising (Rudin–Osher–Fatemi): recover a clean image u from a noisy one g by

$$\min_u \underbrace{\int \|\nabla u\| \, d\mathbf{x}}_{\text{smoothness}} + \frac{\alpha}{2} \underbrace{\int (u - g)^2 \, d\mathbf{x}}_{\text{stay close to data}}.$$

Original



Noisy image



Denoised image



Source: Wikipedia, Rudin et al. 1992.

Generalized Fick's law for c

$$\mathcal{L}\left(\textcolor{blue}{c}, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(\textcolor{red}{c}^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(\textcolor{blue}{c} - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c} = -\frac{\partial \mathcal{L}}{\partial c} = c^3 - c + \alpha(c - \varphi),$$

$\Rightarrow$ 2nd-order PDE, no 4th-order terms

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

Euler-Lagrange equation for φ

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

For φ , we simply write the Euler-Lagrange equations.

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \varphi_t} \right) + \operatorname{div} \left(\frac{\partial \mathcal{L}}{\partial \nabla \varphi} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}.$$

which gives

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div} (\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far, instead of

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c),$$

we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \gamma \Delta \varphi = \alpha(c - \varphi). \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far, instead of

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c),$$

we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \gamma \Delta \varphi = \alpha(c - \varphi). \quad (II)$$

- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.

2nd-order approximation of the Cahn-Hilliard equation

Thus so far, instead of

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c),$$

we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \gamma \Delta \varphi = \alpha(c - \varphi). \quad (II)$$

- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.
- Equation (II) is a hyperbolic wave equation with right-hand side. $\Rightarrow$ Order reduction.

Cattaneo relaxation for equation (I)

$$\begin{aligned}\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \nabla \mu &= -\frac{1}{\tau} \mathbf{q},\end{aligned}$$

- $\tau \ll 1$ is a relaxation time.
- c is still a conserved quantity in this framework.

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div} (\gamma \nabla \varphi) = \alpha (c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

Therefore (II) becomes

$$\begin{aligned} \frac{\partial w}{\partial t} - \operatorname{div}(\gamma \mathbf{p}) &= -\alpha(\varphi - c), \\ \frac{\partial \varphi}{\partial t} &= \frac{1}{\beta} w, \\ \frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w &= 0. \end{aligned}$$

Final system approximating the Cahn-Hilliard equations

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla (c^3 - c + \alpha(c - \varphi)) = -\frac{1}{\tau} \mathbf{q}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} (\gamma \mathbf{p}) = -\alpha(\varphi - c)$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{1}{\beta} w \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\beta} w$$

- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.

Hyperbolicity

System admits a full set of real eigenvalues (with $\alpha > 1$) given by

$$\lambda_{1,2} = \pm \frac{\sqrt{3c^2 - 1 + \alpha}}{\sqrt{\tau}}, \quad \lambda_{3,4} = \pm \frac{\sqrt{\gamma}}{\sqrt{\beta}}, \quad \lambda_{5-9} = 0$$

and a corresponding full set of linearly independent eigenvectors.

$\Rightarrow$ System is strongly hyperbolic.

Lyapunov Functional

Proposition

The proposed hyperbolic Cahn-Hilliard system admits the following Lyapunov functional

$$E = \int_{\mathcal{D}} e(c, \varphi, \mathbf{q}, \mathbf{p}, w) \, d\Omega,$$
$$e(c, \varphi, \mathbf{q}, \mathbf{p}, w) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{\alpha}{2} (c - \varphi)^2 + \frac{1}{2\beta} w^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi}$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w}$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

$$\Rightarrow \frac{\partial e}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial c} \frac{\partial e}{\partial \mathbf{q}} - \frac{\partial e}{\partial \mathbf{p}} \frac{\partial e}{\partial w} \right) = - \left\| \frac{\partial e}{\partial \mathbf{q}} \right\|^2 \leq 0,$$

Numerical methods

In order to solve the model numerically and also compare it with reference solutions, we propose here:

- ① A numerical scheme for the original Cahn-Hilliard equation based on 4th order semi-implicit conservative finite differences
- ② Explicit MUSCL-Hancock for the hyperbolic approximation.

Implicit conservative finite differences for CH

We propose here a semi-implicit conservative in order to solve numerically the original Cahn-Hilliard equations. We rewrite the latter as follows

$$\frac{\partial c}{\partial t} - \operatorname{div}(\mathbf{F}) + \gamma \Delta^2 c = 0$$

where $\mathbf{F}$ is the flux given by

$$\mathbf{F} = \chi(c) \nabla c, \quad \chi(c) = 3c^2 - 1$$

The scheme writes

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta \Delta_h c_{i,j}^{n+1}.$$

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using conservative finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^n (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^n \simeq \frac{1}{12} (7\chi_{i,j}^n - \chi_{i-1,j}^n + 7\chi_{i+1,j}^n - \chi_{i+2,j}^n) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12\Delta x} (15c_{i,j}^{n+1} - 15c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} - c_{i-1,j}^{n+1}) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using conservative finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^n (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^n \simeq \frac{1}{12} (7\chi_{i,j}^n - \chi_{i-1,j}^n + 7\chi_{i+1,j}^n - \chi_{i+2,j}^n) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12\Delta x} (15c_{i,j}^{n+1} - 15c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} - c_{i-1,j}^{n+1}) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

These are 4th order approximations.

Discretization of the bi-Laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{1}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{1}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

Discretization of the bi-Laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{1}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{1}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta\Delta_h c_{i,j}^{n+1}.$$

is then solved using GMRES (Matrix is not symmetric ...)

Numerical method for hyperbolic approximation

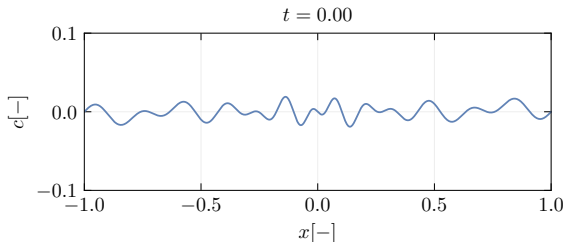
- Explicit second-order MUSCL-Hancock scheme
- We use either FORCE or Rusanov approximate Riemann solvers (One could also implement a Roe solver)

Spinodal decomposition

We suggest the following initial data

$$c(x) = \begin{cases} 0.01 \left((\sin(10\pi(1+x))) - \sin(10\pi(1+x)^2) \right), & \text{if } x \in [-1, 0] \\ -0.01 \left((\sin(10\pi(1-x))) - \sin(10\pi(1-x)^2) \right), & \text{if } x \in [0, 1]. \end{cases}$$

This function is built in such a way that it is C^∞ over $[-1, 1]$ as well as over $\mathbb{R}$ by periodic prolongation.



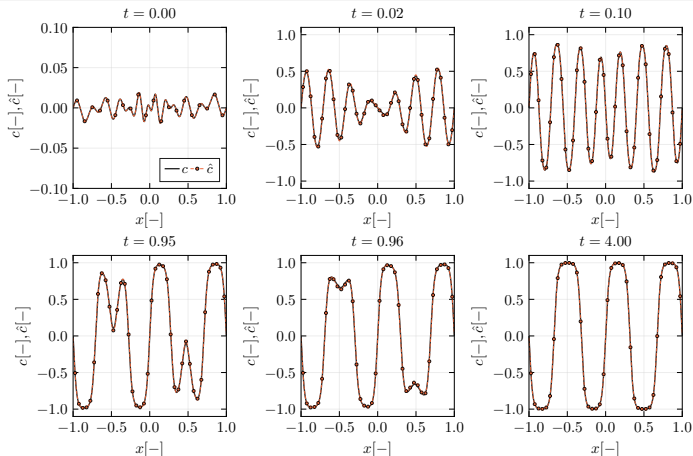
Spinodal decomposition ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

Figure 5: Comparison of the numerical results between the original model (orange) and its hyperbolic counterpart (black). $N = 2000$ computational cells.

Spinodal decomposition - Lyapunov function evolution ($\simeq 6b$ iterations)

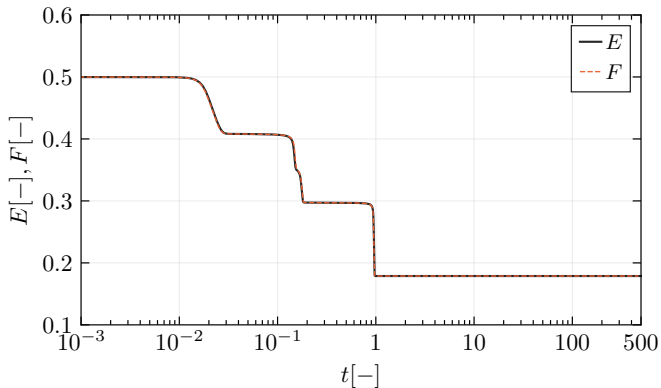


Figure 6: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Lyapunov function evolution ($\simeq 30b$ iterations)

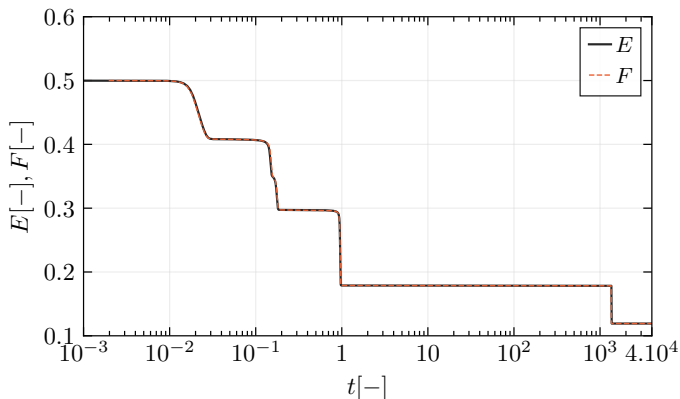


Figure 7: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Ostwald Ripening in 1D ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-4}$)

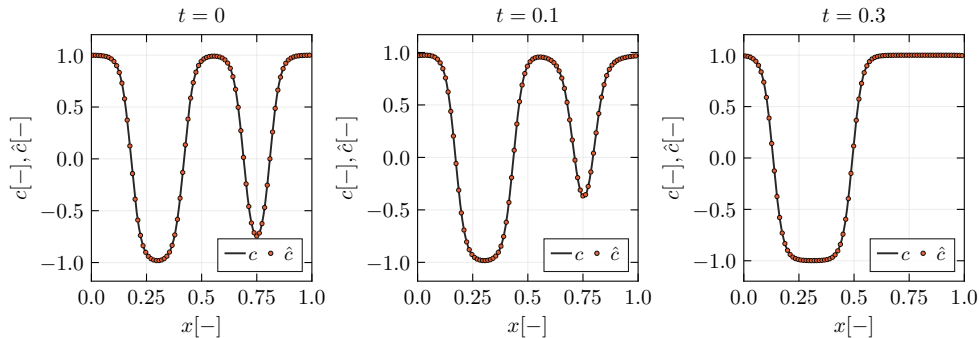
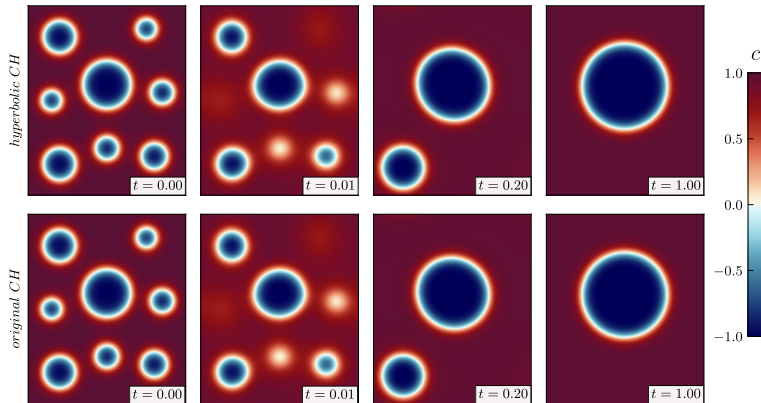


Figure 8: Comparison of the numerical solutions for hyperbolic Cahn–Hilliard model (black line) and the original model (red dots) for the Ostwald Ripening test case at times $t = \{0, 0.1, 0.3\}$.

Ostwald Ripening in 2D($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)



Ostwald Ripening in 2D: horizontal Cuts

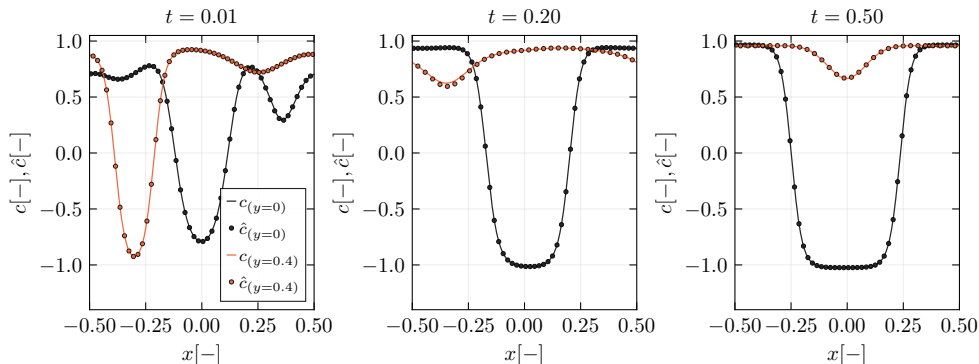


Figure 9: Horizontal cuts over the lines $y = 0$ (red) and $y = 0.4$ (black). Domain is 600×720

Conclusion and Perspectives

Conclusions

- Diffusion equations can be approximated by hyperbolic models.
- Not every hyperbolic approximation conserves properties.
- In both cases, the approach was validated theoretically and numerically using standard numerical methods.

Perspectives

- Is it worth it ? A deeper analysis of ups and downs.
- A more suited numerical machinery: Semi-implicit discretization, asymptotic preserving schemes, adaptive time-stepping, etc ...
- More complex models, couplings with Navier-Stokes equations, etc.

Thank you for your attention !

[1] Dhaouadi, Firas, and Gavriluk, Sergey. "An Eulerian hyperbolic model for heat transfer derived via Hamilton's principle: analytical and numerical study." Proceedings of the Royal Society A 480.2283 (2024): 20230440.

[2] Dhaouadi, Firas, Dumbser, Michael, & Gavriluk, S. A first-order hyperbolic reformulation of the Cahn–Hilliard equation. " Proceedings of the Royal Society A (Vol. 481, No. 2312, p. 20240606).

And references therein.

Codes published alongside the papera.

Kinetic theory of rarefied gases: Bhatnagar-Gross-Krook (BGK) model

Kinetic model (cf. W. Haegeman, G. Orlando, M. Massot, S. Kokh (2024)):

$$f_t + \mathbf{v} \cdot \nabla_{\mathbf{x}} = \frac{M[f] - f}{\tau}.$$

Eight-moments system

$$\rho_t + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$(\rho \mathbf{u})_t + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = 0,$$

$$(\rho E)_t + \operatorname{div}((\rho E + p) \mathbf{u} + \mathbf{q}) = 0,$$

$$\begin{aligned} \mathbf{q}_t + (\mathbf{u} \cdot \nabla) \mathbf{q} + \frac{2\gamma - 1}{\gamma} \left(\frac{\partial \mathbf{u}}{\partial x} + \operatorname{div}(\mathbf{u}) \mathbf{I} \right) \mathbf{q} + \frac{\gamma - 1}{\gamma} \left(\frac{\partial \mathbf{u}}{\partial x} \right)^T \mathbf{q} \\ + \frac{\gamma p}{\gamma - 1} \nabla \left(\frac{p}{\rho} \right) = -\frac{\mathbf{q}}{\tau}, \quad \gamma = \frac{5}{3}. \end{aligned}$$

Compound shocks

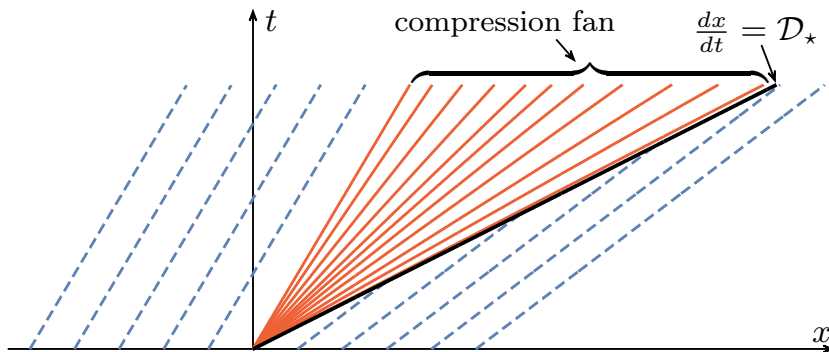


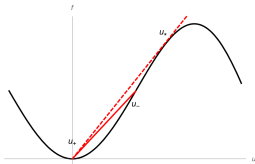
Figure 10: Schematic representation of the wave pattern in the $x - t$ plane, for a compound shock solution. The shock propagates to the right, followed by a right facing compression fan.

Oleinik-Liu admissibility criterion

O. Oleinik (scalar conservation law), B. Wendroff (Euler equations with $p_v < 0$), T.- P. Liu (systems of conservation laws).=====

Scalar conservation laws:

$$u_t + f(u)_x = 0.$$



The shock is *admissible* iff

$$D = \frac{f(u_-) - f(u_+)}{u_- - u_+} \geq \frac{f(u) - f(u_+)}{u - u_+}$$

for any u in the interval (u_+, u_-) .

Admissible states

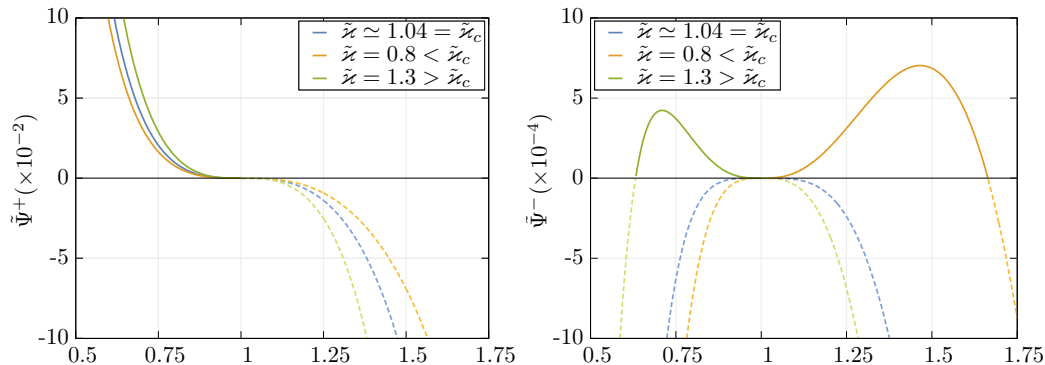


Figure 11: Plot of $\tilde{\Psi}^+$ (Left) and $\tilde{\Psi}^-$ (Right) as a function of $\tilde{v}$ for different values of $\tilde{\kappa}$ that are $\kappa_c \simeq 1.0399$, $\tilde{\kappa} = 0.8$ and $\tilde{\kappa} = 1.3$. $\gamma = 2$. The dashed lines correspond to non-physical solutions that do not satisfy $\tilde{\Psi} \geq 0$.